Wei-Chen (Gordon) Kuo

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Education

National Cheng Kung University Sep. 2021 – Jun. 2026
Bachelor of Arts in Political Science *Tainan, Taiwan*
Minor in Transportation & Communication Management Science
GPA: 3.41/4.30
Athletic Scholarship (Basketball): 50% tuition scholarship for two semesters; sole university-wide recipient.
Selected Graduate-level Coursework: *Operations Research (B-), Mathematical Programming (A-), Optimization Models and Applications (A), Supply Chain Optimization (A+)*

Work Experience

Micron Technology Taiwan Inc. Expected Dec. 2026
Incoming Engineer, F11 ME FST WH (Manufacturing, Materials, Grade: E1) *Taoyuan, Taiwan*

ASML Taiwan Ltd. Jun. 2025 – Jun. 2026
Industrial Engineer Intern (Manufacturing, Business Engineering) *Tainan, Taiwan*
Continuous Improvement MVP, Oct. 2025 — recognized for contributions to four continuous improvement projects; sole intern factory-wide recipient

Manufacturing Decision Support

- Developed a Python scheduling decision-support tool to identify and correct mis-scheduled work orders that inflated machine idle time, preventing capacity plans from overstating equipment requirements by two machines.
- Developed a reusable multi-period capacity-planning model that translates production demand into period-specific FTE, equipment, facility space, and capacity requirements while identifying resource shortages; applied it to 2025–2029 orders and reduced computation time by 83% (∼40 minutes to ∼5 minutes).
- Developed a factory-wide machine-runtime monitoring dashboard with automated idle-equipment detection and engineer notifications, enabling continuous monitoring of 24/7 manufacturing operations.

Workflow Automation & Operations Analytics

- Developed a Python-based STO planning system to determine weekly transfer quantities across warehouse and factory locations from inventory, demand, and material-shortage data, reducing processing time by 98% (∼30 minutes to ∼30 seconds).
- Developed an end-to-end Power Automate workflow supporting 100+ engineers, automating training registration, task assignment, notifications, status tracking, and SharePoint recordkeeping.

IMS Nanofabrication GmbH Apr. 2025 – Jun. 2025
Logistics & Service Support Intern *Tainan, Taiwan*

- Coordinated end-to-end parts-exchange workflows between warehouse and fabrication operations to support semiconductor service activities and TSMC customer operations.
- Analyzed logistics workflows and assisted with on-site problem-solving within fabrication facilities, contributing to process-improvement initiatives and operational efficiency.

Research Experience

Undergraduate Thesis / Unpublished Manuscript 2026
Beyond Terminal Punctuality: Evaluating the Effect of Railway Corporatization in Taiwan, Advisor: Pei-Chun Lin

- Evaluated TRC punctuality following its 2024 corporatization using monthly 2021–2025 data and a benchmark difference-in-differences comparison with THSRC; estimated a 1.50 percentage-point relative decline in official punctuality.
- Assessed robustness using Newey–West HAC standard errors, false-treatment-date placebo tests, and leave-one-month-out sensitivity analysis; coefficient estimates remained negative and pre-treatment placebo effects were insignificant.
- Constructed a station-stop-exposed punctuality measure and identified a 14.4 percentage-point gap between TRC's official terminal-based punctuality rate (94.7%) and the exposure-based measure (80.3%), highlighting intermediate-station delays omitted by terminal-only reporting.

Global Logistics Management Research Center, NCKU Jul. 2025 – Jul. 2026

Research Assistant, advisor: Pei-Chun Lin

- Contributed to the study design for an ongoing patient-allocation project using confidential NCKU Hospital data, examining patient allocation between general and forthcoming elderly-care hospitals under equity and financial considerations.
- Conducted descriptive statistical analysis and data visualization to characterize patient and hospital data and inform subsequent development of the optimization framework.
- Contributed statistical modeling and computational analysis to the SSRN working paper Impact of Austria's National Policies on Nursing Workforce: A Synthetic Control Approach (under review), implementing synthetic control analysis to evaluate national policy effects on nursing-workforce outcomes.

Department of Political Science, NCKU Aug. 2024 – Nov. 2024

Research Assistant, advisor: Issac S. H. Huang

- Processed and structured a 240-document corpus of parliamentary interpellations in Python for the government-funded project Trajectory of Democratic Progress: A Textual Analysis of Taiwan's Parliamentary Interpellations during the Authoritarian Period.

Selected Independent Optimization Projects

Shift Scheduling Optimization for Safety Audit Sep. 2025 – Jan. 2026

Project Supervisor: I-Lin Wang

- Formulated a MILP for safety-audit scheduling and auditor assignment under staffing, qualification, workload, and pairing constraints, with endogenous efficiency gains for inexperienced auditors through pairings with experienced staff.
- Linearized bilinear assignment and learning terms for Python/Gurobi implementation; verified the formulation on a small instance and evaluated a 20-day case with 8 experienced and 14 inexperienced auditors.

School Bus Routing Optimization Sep. 2025 – Jan. 2026

Project Supervisor: I-Lin Wang

- Formulated a MILP to jointly determine school-bus routes, fleet size, and starting stops using a virtual-source formulation for endogenous start selection, with capacity, fleet-cost, route-duration, and student in-vehicle-time considerations.
- Implemented the model in Python/Gurobi with MTZ subtour-elimination and time-propagation constraints; verified optimality on a 10-stop instance and evaluated a 115-stop network, where the solver time limit indicated scalability limits of the exact formulation.

AGV Routing Optimization for Factory Material Delivery Sep. 2025 – Jan. 2026

Project Supervisor: Pei-Chun Lin

- Formulated AGV material-delivery routing as a CVRP, solving a small instance exactly in LINGO and implementing a nearest-neighbor heuristic in Python for a 150-machine network using real warehouse-location data as proxy machine coordinates.

Leadership & Service

Hult Prize Taiwan Feb. 2025 – Jun. 2025

Consumption Team Leader *Tainan, Taiwan*

- Led material-demand forecasting, procurement, inventory planning, and sponsor coordination for Taiwan's first nationwide Hult Prize business competition, serving approximately 100 participants.

NCKU Office of International Affairs Aug. 2024 – Dec. 2024

Campus Linker / Exchange Student Assistant *Tainan, Taiwan*

- Supported international exchange students with university registration, course enrollment, housing, insurance, orientation, and cross-cultural adaptation.

NCKU Men's Basketball Team Jul. 2023 – Jul. 2024

Captain *Tainan, Taiwan*

- Led the university men's basketball team, managing training, competition schedules, finances, and equipment while organizing major athletic events, including the UBA Southern Regional Preliminaries.

Skills

Programming: Python, R, Visual Basic for Applications (VBA)

Optimization & Statistical Software: Gurobi, LINGO, SPSS, GAUSS

Data & Automation Tools: Excel, Power BI, Power Automate, Microsoft SharePoint

Methods: Mixed-Integer Linear Programming, Vehicle Routing, Scheduling, Difference-in-Differences, Synthetic Control

Languages: Chinese (Native), English (TOEFL iBT - 5.5/6, equivalent to 110/120), German (Basic)